

Unit 1

Numerical Reasoning

Fun Puzzle 1:

[https://transum.org/Software/
SW/Starter_of_the_day/starter
_June1.asp](https://transum.org/Software/SW/Starter_of_the_day/starter_June1.asp)

Fun Puzzles 2:

[https://transum.org/Software/
SW/Starter_of_the_day/starter
_June2.asp](https://transum.org/Software/SW/Starter_of_the_day/starter_June2.asp)

Fun Puzzles 3:

**[https://transum.org/Software/
SW/Starter_of_the_day/starter
_June3.asp](https://transum.org/Software/SW/Starter_of_the_day/starter_June3.asp)**

Tangram Builder:

<https://polypad.amplify.com/tangram>



Puzzle 4:

https://www.transum.org/Maths/Puzzles/Les_Page/Brainbox/



Unit Details: Numerical Reasoning

Specified Concept: Representation & Systems

Inquiry Statement: How have number systems changed over time, and why did different civilizations create their own ways to represent numbers?

Global Context Focus: Orientation in space and time.

ATL focus: Research and Communication Skills



Reflections before the unit: *[Discussion with students]*

1. Where do you see numbers being used most often outside the mathematics classroom, and why do you think people rely on them?



Reflections before the unit: *[Discussion with students]*

2. Can two people look at the same numerical information and reach different conclusions? Give an example.



Reflections before the unit: *[Discussion with students]*

3. If numbers suddenly disappeared for one day, what activities in your life would become difficult?



Reflections before the unit: *[Discussion with students]*

4. Which number representation do you encounter most often—fractions, decimals, or percentages? Why?



Reflections before the unit: *[Discussion with students]*

5. Are all numbers equally useful in every situation? Explain your thinking.



Week 1 Content:

- Forms of numbers
- Fractions, decimals, and percentages
- Conversions between fractions, decimals, and percentages
- Comparing and ordering numbers in different forms



Situation		Number
Sale in a store		25% OFF
Exam result		0.75
Pizza shared		$\frac{3}{4}$
Bank interest		5%
Fuel tank full		1.00



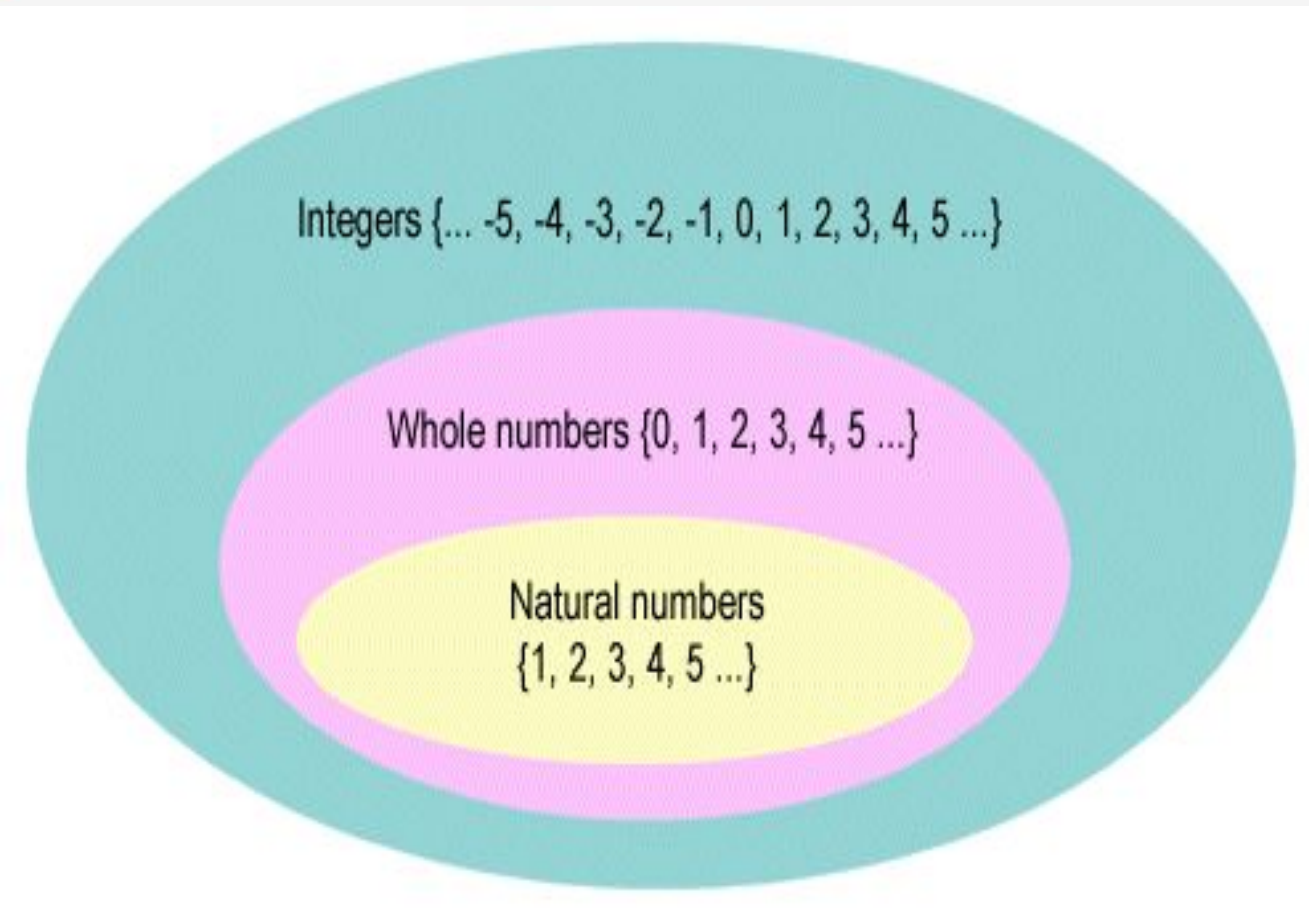
Think – Pair – Share

[Students think about the answer – Pair up and then share their points with each other and then with the teacher]

Questions:

1. Are these numbers the same type?
2. Where have you seen them?
3. Why do people use different forms?
4. In the previous slides [Is the fraction with the pizza is correct, reason out]



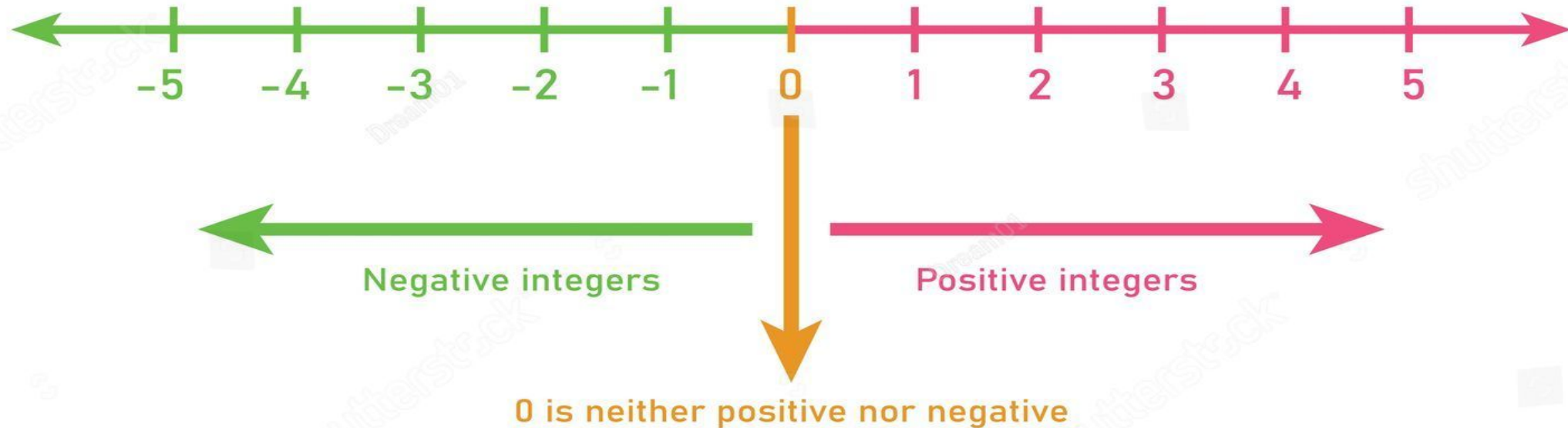


1. What is Number & Digit?
2. Do you see any numbers that are missing from one group but present in another?
3. Why do you think 0 is not in Natural Numbers?
4. Why might people need negative numbers in real life?
5. Why do you think mathematicians created different number sets?

<https://docs.google.com/document/d/16jlw0fI9dGIGuAMrWJlpiC79nsoMow-63R3LkKUU3o/edit?tab=t.0>

Integers Game

Integers on Number Line



INTEGERS (Solve the problems on the number line)

1. $+2-3$

2. $-2-3$

3. $+3-2$

4. $+2+3$

5. $+4-5$

6. $-5-4$

7. $+4+5$

8. $+5-4$

9. $+6-1$

10. $-6-1$

11. $+6+1$

12. $-6+1$

13. $+7+8$

14. $+8-7$

15. $-8+7$

16. $-7-8$



Investigate the number-line patterns and develop your own addition and subtraction rules for integers.



INTEGERS

ADDING INTEGERS

+ SAME -
+ SIGNS -

1. Add the absolute values of the numbers.

2. Keep the common sign.

$$-3 + (-5) = -8$$

$$3 + 5 = 8$$

DIFFERENT
- SIGNS +

1. Subtract the absolute values of the numbers.

2. Take the sign of the greater absolute value.

$$3 + (-5) = -2$$

$$-3 + 5 = 2$$

SUBTRACTING INTEGERS

1. Add the opposite.

$$-3 - (-5)$$

$$-3 + 5$$

2. Follow the addition rules.

MULTIPLYING & DIVIDING INTEGERS

+ SAME -
+ SIGNS -

Answer is positive

$$+ \div + = +$$

$$- \cdot - = +$$

$$-3 \cdot (-3) = 9$$

DIFFERENT
- SIGNS +

Answer is negative

$$- \div + = -$$

$$- \cdot + = -$$

$$8 \div (-2) = -4$$

+ 1. Addition Rules

Rule	Example
Same signs → Add and keep the sign	$(-3) + (-4) = -7$ $(+5) + (+2) = +7$
Different signs → Subtract and take the sign of the bigger number	$(-6) + (+2) = -4$ $(+7) + (-3) = +4$

— 2. Subtraction Rules

💡 **Tip:** Change subtraction to addition by taking the **opposite** of the second number.

$$a - b = a + (-b)$$

Example	Converted	Answer
$5 - 3$	$5 + (-3)$	$= 2$
$-4 - 2$	$-4 + (-2)$	$= -6$
$-2 - (-5)$	$-2 + 5$	$= 3$



✖ 3. Multiplication Rules

Sign of Numbers	Result Sign	Example
Same signs	Positive (+)	$(-3) \times (-2) = +6$
Different signs	Negative (-)	$(-4) \times (+5) = -20$
Any number $\times 0$	0	$7 \times 0 = 0$

÷ 4. Division Rules

Sign of Numbers	Result Sign	Example
Same signs	Positive (+)	$(-12) \div (-3) = +4$
Different signs	Negative (-)	$(+10) \div (-2) = -5$



Write the correct answer for each question

① $(-5) + 7 = \underline{\hspace{2cm}}$

② $(-27) + (-12) = \underline{\hspace{2cm}}$

③ $27 - (-2) = \underline{\hspace{2cm}}$

④ $14 + (-8) = \underline{\hspace{2cm}}$

⑤ $(-9) + (2) = \underline{\hspace{2cm}}$

⑥ $33 + 9 = \underline{\hspace{2cm}}$

⑦ $6 + (-9) = \underline{\hspace{2cm}}$

⑧ $28 - 3 = \underline{\hspace{2cm}}$

⑨ $(-10) + (-2) = \underline{\hspace{2cm}}$

⑩ $-9 + 13 = \underline{\hspace{2cm}}$

⑪ $(-5) + (-10) = \underline{\hspace{2cm}}$

⑫ $18 - (-2) = \underline{\hspace{2cm}}$

⑬ $(-9) + (-4) = \underline{\hspace{2cm}}$

⑭ $(-6) + (-9) = \underline{\hspace{2cm}}$



Integers worksheets

[Integers worksheet 1](#)

[Integers worksheet 2](#)

[Integers worksheet 3](#)

[Integer worksheet 4](#)



Integers worksheet 5

Integers worksheet 6

Integers worksheet 6

https://drive.google.com/file/d/1fD7dJSc9groAXMLlxT-jqwRKejMnsbSN/view?usp=drive_link



Word problems on Integers

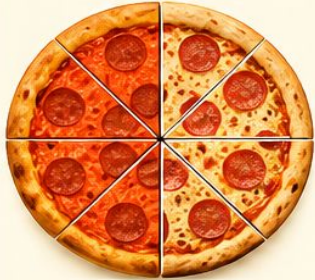


SEE-THINK-WONDER

EXPLORING FRACTIONS, DECIMALS & PERCENTAGES

Students will explore the relationship between fractions, decimals, and percentages using real-life examples.

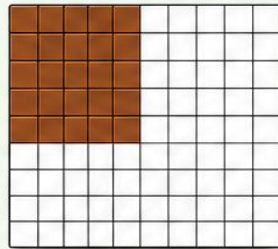
1. PIZZA



4 out of 8 slices are eaten.



2. CHOCOLATE BAR



25 out of 100 squares are eaten.



3. PROGRESS BAR

Project Completed



75%

The project is 75% complete.



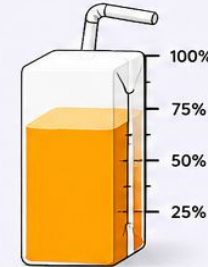
4. SHOPPING DISCOUNT



The item is 50% off.



5. JUICE BOX



The juice box is 75% full.



HOW CAN WE WRITE EACH EXAMPLE AS A FRACTION, DECIMAL AND PERCENTAGE?

EXAMPLE	FRACTION	DECIMAL	PERCENTAGE	IN WORDS
 Pizza (4 out of 8)	$\frac{4}{8} = \frac{1}{2}$	0.5	50%	Fifty percent
 Chocolate (25 out of 100)	$\frac{25}{100} = \frac{1}{4}$	0.25	25%	Twenty-five percent
 Progress Bar (75%)	$\frac{75}{100} = \frac{3}{4}$	0.75	75%	Seventy-five percent
 Shopping Discount (50% off)	$\frac{50}{100} = \frac{1}{2}$	0.5	50%	Fifty percent
 Juice Box (75% full)	$\frac{75}{100} = \frac{3}{4}$	0.75	75%	Seventy-five percent

THINK ABOUT IT!



- How are fractions, decimals, and percentages related?
- Where do we see percentages in real life?
- Which form is easier to understand in daily life? Why?



SEE-THINK-WONDER ACTIVITY

<https://docs.google.com/document/d/1o6zz3r3T-ht87CjYVJl0hys1vW0STMeWvRWIDXBzmKg/edit?tab=t.0>



Criteria B Investigation Task (HOMETASK)

Task: Choose one real-life example (pizza, shopping discount, sports statistics, exam score, or mobile battery).

Investigate and answer:

1. What is the fraction?
2. What is the decimal?
3. What is the percentage?
4. Explain how you converted between the three forms.
5. What pattern did you observe?
6. Which representation is most useful for this situation? Why?

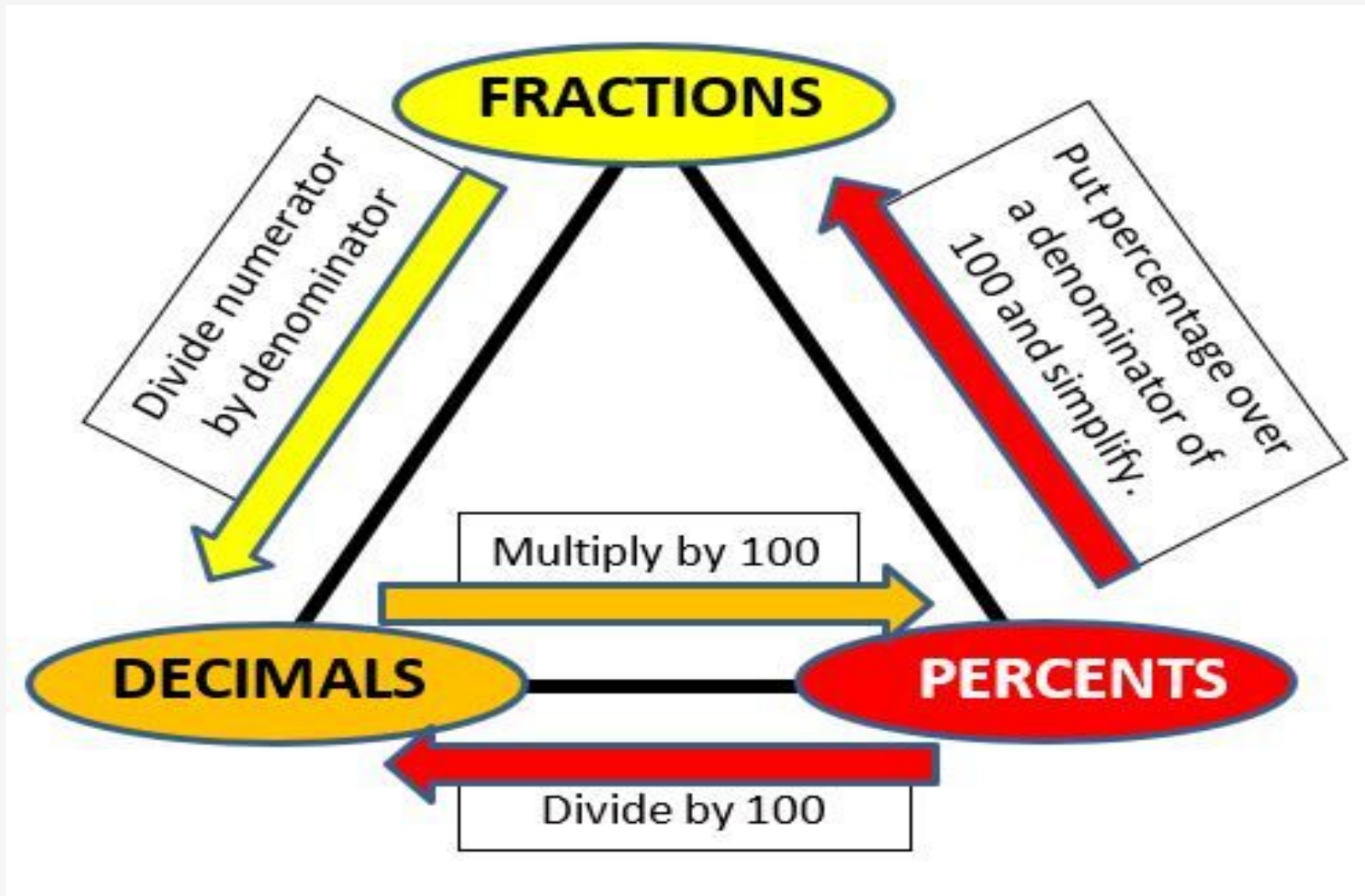


Math Assessment Criteria

https://www.bloomfield.org/uploaded/Parents_Students/IB/myp_rubrics/Math_Year_1_Rubrics.pdf



Relationship between fractions - decimals and percentages



Converting fractions - decimals and percentages

Decimal	Percentage	Fraction
0.23	23%	$\frac{23}{100}$ cannot simplify!
$\times 100$	out of 100	
0.05	5%	$\frac{5}{100} \rightarrow \frac{1}{20}$ can simplify!
$\times 100$	out of 100	

Percentage	Decimal	Fraction
7%	0.07	$\frac{7}{100}$ cannot simplify!
$\div 100$	out of 100	
12%	0.12	$\frac{12}{100} \rightarrow \frac{3}{25}$ can simplify!
$\div 100$	out of 100	



Percentage Art Activity Using Graph Paper:

<https://docs.google.com/document/d/1cf99l0zKZywjzVkgZBvaQpqRI1WnMBzLLTJ8pJjMDr4/edit?tab=t.0#heading=h.dlzvnwn7pcza>

Word wall on conversions

<https://wordwall.net/resource/92013921>

Worksheet -

<https://www.k5learning.com/worksheets/math/grade-6-fractions-decimals-percents-a.pdf>





Practice 3

1 Convert each decimal to a fraction and then a percentage.

a $0.87 = \frac{\square}{\square} = \square\%$ **b** $0.5 = \frac{\square}{\square} = \square\%$ **c** $0.732 = \frac{\square}{\square} = \square\%$

d $0.09 = \frac{\square}{\square} = \square\%$ **e** $0.006 = \frac{\square}{\square} = \square\%$ **f** $1.2 = \frac{\square}{\square} = \square\%$

2 Convert each percentage to a decimal.

a 12% **b** 46% **c** 103% **d** 38.5% **e** 8.3% **f** 12.25%

3 Write these numbers in all three forms: fraction, decimal and percentage.
Write each fraction in its simplest form.

a $\frac{2}{25}$

b 0.8

c 35%

d $\frac{4}{9}$



Taken from - Oxford textbook pg :64

Weekend Task

https://docs.google.com/document/d/1CvNuudAw5pvs8XtSdZYN13SV_uaeByVx/edit



Decimal Recap

Equivalent Fractions practice



- Q**
- a** What percentage of 80 is 8?
 - b** What is 30% of 25?
 - c** 50% of what number is 13?

A **a** 80 is the whole and 8 is the part, so the fraction is $\frac{8}{80}$.

$$\begin{aligned}\frac{8}{80} &= \frac{8 \div 8}{80 \div 8} = \frac{1}{10} \\ &= \frac{1 \times 10}{10 \times 10} = \frac{10}{100} = 10\%\end{aligned}$$

Therefore, 8 is 10% of 80.

b $\frac{30}{100} = \frac{\square}{25}$

$$\frac{30 \div 4}{100 \div 4} = \frac{7.5}{25}$$

Therefore, 30% of 25 is 7.5.

c $\frac{50}{100} = \frac{13}{\square}$

$$\begin{aligned}\frac{1}{2} &= \frac{13}{\square} \\ \frac{1 \times 13}{2 \times 13} &= \frac{13}{26}\end{aligned}$$

Therefore, 50% of 26 is 13.

Convert the fraction to a percentage.

In this case, 25 is the whole and the part is unknown.
We need to find the equivalent to the fraction represented by 30%.

Since $100 \div 4 = 25$, we need to divide the numerator by 4 as well.

In this case, 13 is the part but the whole is unknown.
We need to find the equivalent to the fraction represented by 50%.

Simplify the first fraction.



Practice 4

1 Find the percentage in each of the following amounts.

- | | |
|--|---|
| a What percentage of 100 is 75? | b What percentage of 48 is 36? |
| c What percentage of 300 is 75? | d What percentage of 620 is 124? |
| e What percentage of 50 is 18? | f What percentage of 220 is 132? |
| g What percentage of 96 is 24? | h What percentage of 850 is 340? |

2 Find the number when:

- | | |
|-----------------------------------|------------------------------------|
| a 50% of the number is 20 | b 20% of the number is 8 |
| c 10% of the number is 12 | d 25% of the number is 38 |
| e 80% of the number is 208 | f 14% of the number is 49 |
| g 62% of the number is 124 | h 130% of the number is 78. |

3 Find the number when given the percentage.

- | | | |
|---------------------|----------------------|---------------------|
| a 25% of 100 | b 40% of 100 | c 20% of 50 |
| d 80% of 100 | e 10% of 1000 | f 70% of 200 |
| g 15% of 200 | h 5% of 60 | i 100% of 80 |

Taken from –
Oxford
textbook pg
68 & 69



Fraction, Decimal and Percents



Week 3 Content:

- Fractions, decimals, and percentages
- Basic Profit & loss
- Properties of Numbers



I bought this chocolate
for ₹20.



- 1) I bought this chocolate for ₹20. If I sell it to my friend for ₹25, what happened?"
- 2) What if I sold it for only ₹15?



Bought → Cost Price (CP)

Sold → Selling Price (SP)

Real-Life Examples

Example 1: A shopkeeper buys a water bottle for ₹80. He sells it for ₹100.

- Who bought it?
- Who sold it?
- Did he gain or lose money?

Example 2: A fruit seller buys apples for ₹200.

- Some apples spoil.
- He sells them for only ₹170.

Example 3: A student buys a used book for ₹150. Later sells it for ₹180.



Class Activity (Market Game)

Item	Cost Price	Selling Price
Pen	₹ 10	₹ 12
Notebook	₹ 50	₹ 45
Football	₹ 500	₹ 550
Juice	₹ 30	₹ 35
Bag	₹ 700	₹ 820



WELCOME TO OUR NEW TOPIC!

PROFIT AND LOSS

Have you ever bought something and sold it?



Do we always make more money?



In real life, when we buy something and sell it, we can make a **PROFIT** or have a **LOSS**.

WHAT IS IT?

- **Cost Price (CP):** The price at which we buy an item.
- **Selling Price (SP):** The price at which we sell an item.
- **Profit:** When SP is more than CP.
- **Loss:** When SP is less than CP.

EXAMPLES FROM REAL LIFE

PROFIT



Bought a pen for ₹10
Sold it for ₹15
Profit = ₹5

PROFIT



Bought a book for ₹120
Sold it for ₹150
Profit = ₹30

LOSS



Bought apples for ₹200
Sold them for ₹170
Loss = ₹30

WHY IS IT IMPORTANT?



Helps us make smart buying and selling decisions.



Shops and businesses need profit to grow.



Helps us understand discounts and sales.



Useful in our daily life and future too!

Think about it!



Can a shopkeeper sell an item for less than the price he bought it?

PROFIT



$SP > CP$
We gain money.



LOSS

$SP < CP$
We lose money.

LET'S DISCUSS!



- Why do people sell things?
- Can we always make profit?
- Where do we see profit and loss around us?



Today we start our journey to become **SMART MATHEMATICIANS** and **SMART DECISION MAKERS!**



Probing Questions

- Why do shopkeepers sell products?
- Can a business always make a profit?
- Why might someone sell an item for less than they bought it?
- Have your parents ever waited for a sale before buying something?
- Why do some products become cheaper over time?
- Can customers benefit when shops make less profit?
- Which is better for a business—always making a huge profit or keeping customers happy?



Exit Ticket

1. What is Cost Price?
2. What is Selling Price?
3. Give one example of profit from your home.
4. Give one example of loss.

Homework

Find three items at home.

Ask your parent:

- Approximately how much was it bought for?
- If sold today, would it make a profit or loss?



If SP is greater than CP

$$Profit = SP - CP$$

If CP is greater than SP

$$Then Loss = CP - SP$$

If Profit and Loss are in rupees

$$Then Profit Percent = \frac{PROFIT}{COST PRICE} \times 100$$

$$or P\% = \frac{P}{CP} \times 100$$

$$Loss Percent = \frac{LOSS}{COST PRICE} \times 100$$

$$or \frac{L}{CP} \times 100$$



Calculate Profit or Loss Problems

1 A toy store bought a toy car for ₹250 and sold it for ₹320.

Find: Profit, Cost Price and Selling Price

2. Rahul bought a mobile phone for ₹15,000. After one year, he sold it for ₹12,500.

Find: Loss and Give two reasons why mobile phones lose value.

3. A family bought a bicycle for ₹6,000. Two years later, they sold it for ₹4,800.

Find: Loss and Why do bicycles usually lose value over time?



Calculate Profit or Loss percentage Problems



"True or False?"

- a. $8 + 5 = 5 + 8$
- b. $7 \times 4 = 4 \times 7$
- c. $(3 + 5) + 2 = 3 + (5 + 2)$
- d. $(2 \times 6) \times 5 = 2 \times (6 \times 5)$
- e. $12 + 0 = 12$
- f. $9 \times 1 = 9$
- g. $8 \times (3 + 2) = (8 \times 3) + (8 \times 2)$

"What do you notice?"



Activity 1 – Pattern Detectives

<https://docs.google.com/document/d/102QX0gbKQv6LiGknMKyjjqHmmbRMD5A10BFA3mSLpN8/edit?tab=t.0>

Activity 2 – Guess the Property

https://docs.google.com/document/d/1_zVIVgbHB92Q3LkljcX0ia_e3bbj801z1XEaPZMxAW8/edit?tab=t.0#heading=h.z60hdn5qhw6c



WEEK 4

- Factors and Multiples
- Prime Numbers.
- Prime Factorization
- LCM & HCF



Activity

Hook Activity (Number Detective)

<https://docs.google.com/document/d/1BGMuNrviC9CIEANuw4mTsb2WeUiyn2p4qLUI7w9iN0/edit?tab=t.0#heading=h.2f3hlh7x8w8q>



Factors & Multiples:

Factors

"Factors" are the numbers we can **multiply together** to get another number:

$$\begin{array}{ccc} 2 & \times & 3 = 6 \\ \nearrow & & \nearrow \\ \text{Factor} & & \text{Factor} \end{array}$$

2 and 3 are factors of 6

A number can have **many** factors.

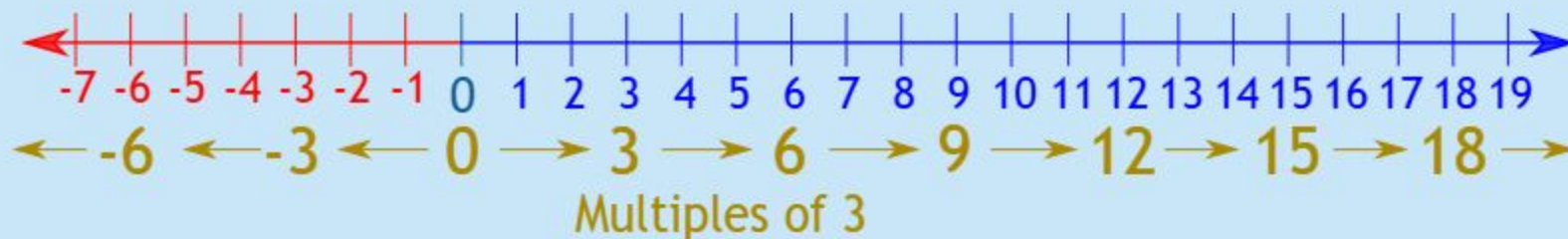


Multiples

A multiple is the result of **multiplying** a number **by an** **integer** (not a fraction).

Example: Multiples of 3 are

..., -9, -6, -3, 0, 3, 6, 9, ...



Example: 15 **is** a multiple of 3, as $3 \times 5 = 15$

Example: 16 is **not** a multiple of 3

Factors & Multiple Problems:

List the factors of:

a 23

b 24

c 100

d 45

e 64

f 72

List the first 5 multiples of:

a 6

b 11

c 12

d 15

e 20

f 35



Investigation of Prime & Composite numbers:

Number	Factor Pairs	
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		



Home Task:



Investigation 1 – The sieve of Eratosthenes

In this investigation you will determine what Eratosthenes' algorithm does.



	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50
51	52	53	54	55	56	57	58	59	60
61	62	63	64	65	66	67	68	69	70
71	72	73	74	75	76	77	78	79	80
81	82	83	84	85	86	87	88	89	90
91	92	93	94	95	96	97	98	99	100

You will need 4 colors (preferably highlighters) to do this investigation. Start with a grid of numbers from 2 to 100, like the one shown here.

- 1 The first number in the list is 2, but do not highlight the 2. Highlight every 2nd number in the grid *after* 2 (all the multiples of 2).
- 2 At this point, the lowest number not yet highlighted is 3. Do not highlight the 3, but highlight every 3rd number in the grid *after* 3 (all the multiples of 3).
- 3 The next lowest number not yet highlighted is 5, so highlight all the multiples of 5.
- 4 Now the next lowest number not yet highlighted is 7, so highlight all the multiples of 7.
- 5 Take a look at all of the numbers that are not highlighted. Describe the pattern of these numbers. What kind of numbers are they?



Prime & Composite numbers(Class discussion)

A **prime** number is a natural number which has exactly two different factors, 1 and itself.

A **composite** number is a natural number which has more than two factors.

- 1
 - a List all the prime numbers up to 29.
 - b Explain why 1 is not a prime number.
 - c How many even prime numbers are there?
 - d List the prime numbers between:
 - i 30 and 40
 - ii 60 and 70
 - iii 90 and 110.



Prime Factorisation

"Can 12 be written as a multiplication?"

- 3×4
- 2×6
- 1×12

Circle 3×4 .

"Is 3 a prime number?" ✓ Yes

"Is 4 prime?" ✗ No

Break 4 again.

$$4 = 2 \times 2$$

Now write:

$$12 = 3 \times 2 \times 2$$

"Are all these numbers prime?"



Prime Factorisation

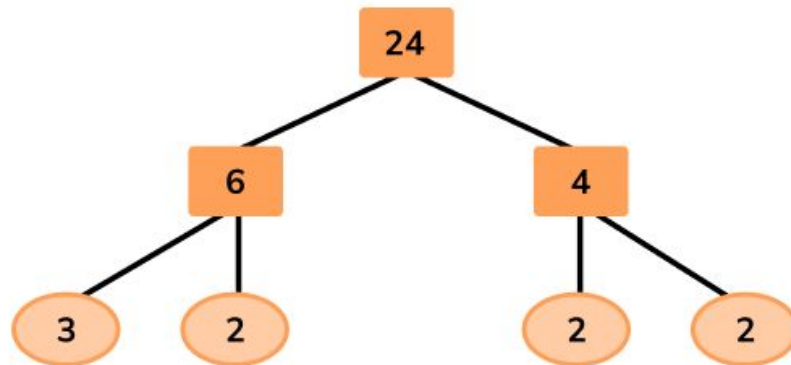
"When we keep breaking a number until only prime numbers remain, it is called **prime factorisation**."

Factor Trees

Factor trees are a way of expressing the factors of a number, specifically the prime factorization of a number.

To do this, list the factors of a number until the number has been reduced to just prime factors.

 **Example** here is the factor tree for the number 24:



$$24 = 2 \times 2 \times 2 \times 3$$

or

$$24 = 2^3 \times 3$$



Prime Factorisation(Ladder Method)

What are the prime factors of:

a 7

b 12

c 50

d 42

e 108

f 210?



(Ladder Method)

Factoring and LCM

Use the Ladder for LCM:

- 1 **WRITE** the two numbers on one line.
- 2 **DRAW THE L SHAPE**
- 3 **DIVIDE** out common prime numbers starting with the smallest (**REPEAT**)

Example:

2	24	36
2	12	18
3	6	9
	2	3



LCM (Least Common Multiple)

Q Use the ladder method to find the LCM of the numbers 4, 8 and 10.

A $2 \mid 4 \quad 8 \quad 10$

Find a factor of *at least* two of the numbers. The more of the numbers your factor divides into, the more efficient the process. Write this factor in front of the three numbers, separated by a vertical line.

$2 \mid 4 \quad 8 \quad 10$
 $2 \quad 2 \quad 5$

Divide each number by the factor (**2**, in this example) and write the quotients below the original numbers.

$2 \mid 4 \quad 8 \quad 10$
 $2 \mid 2 \quad 4 \quad 5$
 $1 \quad 2 \quad 5$

Now find a factor (**2** again, here) of at least two of the new numbers and write it in front. If a number does not have that factor simply carry it down to the next row. (Here, 5 does not have 2 as a factor.)

$2 \mid 4 \quad 8 \quad 10$
 $2 \mid 2 \quad 4 \quad 5$
 $1 \quad 2 \quad 5$

Continue until no more factors can be found. Multiply all the **factors** and **remainders** together to find the lowest common multiple.

$$2 \times 2 \times 1 \times 2 \times 5 = 40$$

The LCM of 4, 8 and 10 is 40.



LCM (Ladder Method)

1 Find the LCM of each set of numbers.

a 4, 6

b 8, 12

c 12, 64

d 24, 32

e 35, 85

f 18, 54, 66

g 16, 20, 80

h 48, 68, 94

i 108, 180, 396



HCF

Q Find the GCF of the numbers 12, 18 and 36.

A 12: 1, 2, 3, 4, **6**, 12
18: 1, 2, 3, **6**, 9, 18
36: 1, 2, 3, 4, **6**, 9, 12, 18, 36

The GCF is **6**.

List all of the factors of each number and find the greatest one that occurs in all the lists.



HCF

Q Using the ladder method, find the GCF of the numbers 12, 18 and 36.

A **3** | 12 18 36

3 | 12 18 36
4 6 12

3 | 12 18 36
2 | 4 6 12

3 | 12 18 36
2 | 4 6 12
2 3 6

$$3 \times 2 = 6$$

The GCF is 6.

Find a factor of *all* the numbers. Write this factor in front of the three numbers, separated by a vertical line. Here, the factor identified is **3**.

Divide each of the original numbers by this factor and write the quotients below the original numbers.

Now find a factor of this new set of numbers and write this in front (**2**). Divide.

Continue until no more factors of *all* the numbers can be found.

Multiply all of the factors together (the numbers to the left of the vertical lines) to find the greatest common factor.



HCF

1 Find the GCF of each set of numbers. Make sure your chosen method is clear.

a 24, 36

b 48, 64

c 54, 81

d 15, 28

e 21, 35, 56

f 120, 56, 96

g 168, 252, 240

h 180, 300, 720

i 75, 200, 350

j 16, 48, 144

k 7, 14, 30

l 18, 28, 38



Solve the following.....

A	Find the highest common factor of 6 and 8	B	Find the lowest common multiple of 4 and 5
C	Find the lowest common multiple of 6 and 8	D	Find the highest common factor of 12 and 20
E	Find the highest common factor of 24 and 40	F	Find the lowest common multiple of 9 and 12
G	Find the lowest common multiple of 20 and 35	H	Find the highest common factor of 72 and 90
I	Find the highest common factor of 75 and 105	J	Find the lowest common multiple of 30 and 105
K	Find the highest common factor of 80 and 128	L	Find the lowest common multiple of 28 and 42
M	Talia works in a café every 8 days. Meg works in the same café every 10 days. How often are they both working in the café at the same time?	N	Hotdog sausages are bought in packs of 9. Hotdog buns are bought in packs of 15. What is the smallest number of hotdogs that can be made with no leftover buns or sausages?

Weekend Task:

https://docs.google.com/document/d/1CvNuudAw5pvs8XtSdZYN13SV_uaeByVx/edit#heading=h.j99y5y6xkpyy



Week 5:

Number Operations

- BODMAS rule.
- Addition, Subtraction of Fraction & Decimal.
- Simplification of fractions.



Task: Unlock using the correct key.



1



34

2



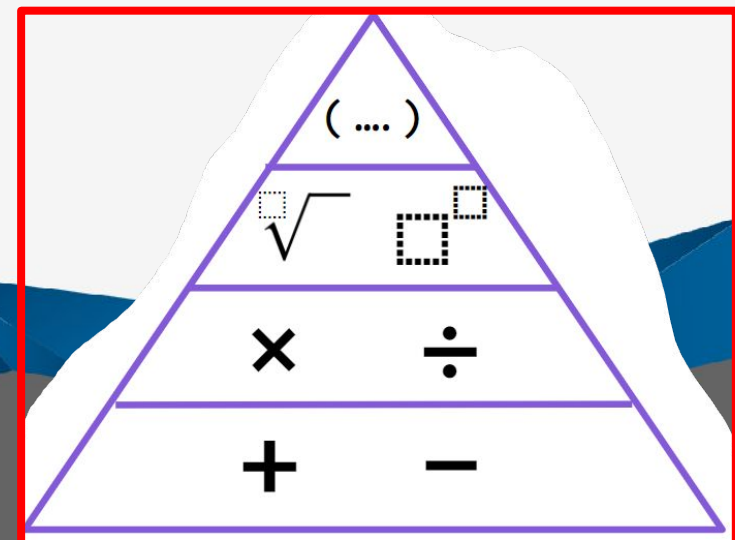
18

CODE: $50 - 6 \times 4 + 8$

Order of Operations

B	Brackets	$10 \times (4 + 2) = 10 \times 6 = 60$
O	Order	$5 + 2^2 = 5 + 4 = 9$
D	Division	$10 \div 6 \div 2 = 10 \div 3 = 13$
M	Multiplication	$10 - 4 \times 2 = 10 - 8 = 2$
A	Addition	$10 \times 4 + 7 = 40 + 7 = 47$
S	Subtraction	$10 \div 2 - 3 = 5 - 3 = 2$

- What do you notice about this expression?
- Which operation would you instinctively perform first? Why?
- How did you decide which operation to perform first?



BODMAS

1 Find the value of each calculation.

a $24 + 62$

b $1 + 8 + 9 + 4 + 2$

c $5 + 3 + 7 + 6 + 5$

d $76 + 83$

e $340 + 220$

f $6 + 8 + 12 + 14 + 20$

g $400 + 1600$

h $98 + 45$

i 10×34.2

j 11×61

k 89×11

l 1230×10

m 8×15

n 10×3.14159

o 15×16

p 4001×10

2 Find the sum of all the whole numbers from 1 to 20 inclusive. Explain your method. Do not add every single number! Think of a strategy to make it easier.

3 Find the sum of all the whole numbers from 1 to 100 inclusive. Explain your method. Do not add every single number! Think of a strategy to make it easier.



Task: Solve the following using BODMAS

$$3 + 5 \times 2$$

$$8 \div 2 \times (3 + 1)$$

$$6 + 3 \times (10 \div 2)$$

$$4 \times (8 - 3) + 2$$

$$20 \div (4 + 1) - 3$$

$$7 - 3 \times (5 - 2)$$

$$5 + 2 \times (8 \div 4)$$

$$12 - 4 \div (2 + 1)$$

$$6 \times (4 - 2) + 5$$

$$24 \div (6 + 2) - 1$$



Order of Operations

Work out:

- (a) $5 + 4 \times 3$
- (b) $4 \times 3 + 5$
- (c) $6 + 10 \div 2$
- (d) $17 + 20 \div 4$
- (e) $30 - 10 \times 2$
- (f) $30 - 20 \div 4$

Order of Operations

Work out:

- (a) $5 + 4 \times 3$
- (b) $4 \times 3 + 5$
- (c) $6 + 10 \div 2$
- (d) $17 + 20 \div 4$
- (e) $30 - 10 \times 2$
- (f) $30 - 20 \div 4$

Add brackets so each calculation is true:

- (a) $20 - 5 \times 3 = 45$
- (b) $5 + 4 - 2 \times 6 = 17$
- (c) $12 + 36 \div 6 - 2 = 21$

Add brackets so each calculation is true:

- (a) $20 - 5 \times 3 = 45$
- (b) $5 + 4 - 2 \times 6 = 17$
- (c) $12 + 36 \div 6 - 2 = 21$



Replace each $\square$ with an operation to make the statement true.

a $1 + 15 \square 3 = 6$

b $(18 \square 2) \div 10 = 2$

c $15 \square 3 + 2 \square 5 = 15$

Insert brackets where appropriate to make each statement true.

a $18 - 6 \times 3 + 2 = 38$

b $8 + 4 \div 2 + 2 = 3$

c $5 + 3 \times 6 - 10 = 38$

d $16 \square 8 + 4 \square 4 = 3$

e $12 \square 8 - 5 \square 4 = 1$

f $5 \times (3 + \square) \square 6 = 9$

Change just one operation in each statement to make it true.

a $7 \times 3 + 4 \times 5 = 1$

b $(12 - 6) \times 3 + 10 \div 5 = 4$

c $24 - (10 - 6) + 2 \times 5 = 16$

d $12 + 36 + 9 \div 4 = 13$

e $6 \times 8 - 12 \times 2 + 7 = 15$

f $14 + 10 - 5 - 12 - 3 = 15$

In Canada it is illegal to be awarded a prize for a 'game of chance'. So, when companies have contests, winners first have to answer a skill-testing question in order to claim their prize. One company's skill-testing question was to find the answer to this calculation:

$$100 + 3 \div 2 \times 5 + 6 \times 10 \div 2$$

a Find the correct answer.

b Find four other answers to this question that respondents are likely to give. Explain the errors that might be made.

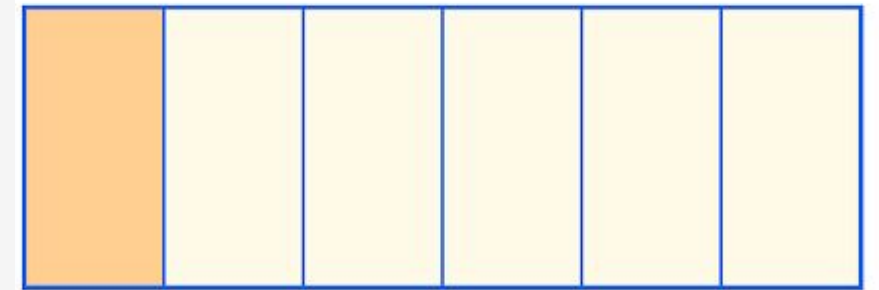
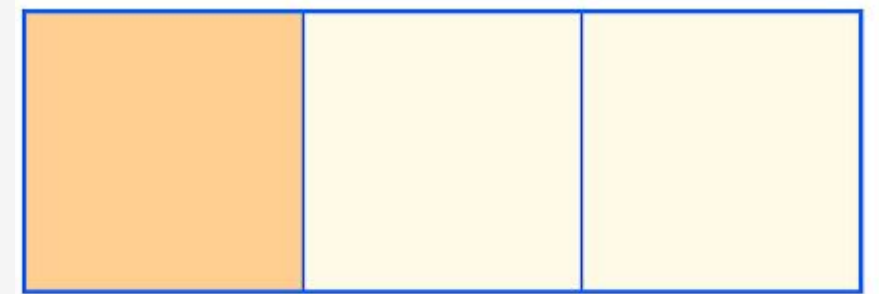


Reflect:

- Can you simply join the shaded parts?
- Are the pieces the same size?
- How can we make all the pieces equal before adding them?
- How can we represent these in fractional form?

$$\frac{1}{3} + \frac{1}{6}$$

- Can we simply add the numerators and denominators?
- If not, what mathematical rule or convention might we need this time?
- Common Denominator



Addition & Subtraction of Unlike Fractions



$$\frac{1}{3} + \frac{1}{6}$$

Convert into like fractions:

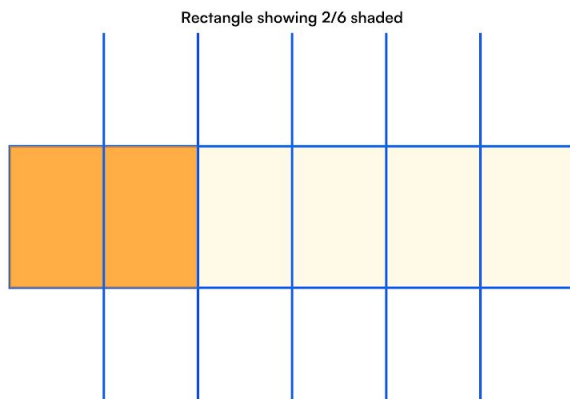
The LCM of 3 and 6 is 6.

$$- \frac{1}{3} = \frac{1 \times 2}{3 \times 2} = \frac{2}{6}$$

$$- \frac{1}{6} \text{ remains } \frac{1}{6}$$

So, the like fractions are $\frac{2}{6}$ and $\frac{1}{6}$.

You can show this by dividing the first rectangle (for $\frac{1}{3}$) into 6 equal parts (by drawing one more line in each third), and shading 2 out of 6 parts.



Thus, $\frac{1}{3}$ and $\frac{1}{6}$ are converted to like fractions $\frac{2}{6}$ and $\frac{1}{6}$.

$$\frac{2}{6} + \frac{1}{6} = \frac{3}{6} = \frac{1}{2}$$



Adding and Subtracting Fractions

(a) $\frac{1}{3} + \frac{1}{4}$

(b) $\frac{1}{3} + \frac{2}{7}$

(c) $\frac{1}{2} + \frac{2}{9}$

(d) $\frac{3}{4} + \frac{1}{5}$

(e) $\frac{1}{3} + \frac{4}{9}$

(f) $\frac{1}{6} + \frac{3}{4}$

(a) $\frac{1}{3} + \frac{1}{4}$

(b) $\frac{1}{3} + \frac{2}{7}$

(c) $\frac{1}{2} + \frac{2}{9}$

(d) $\frac{3}{4} + \frac{1}{5}$

(e) $\frac{1}{3} + \frac{4}{9}$

(f) $\frac{1}{6} + \frac{3}{4}$

(a) $\frac{1}{3} - \frac{1}{5}$

(b) $\frac{1}{3} - \frac{2}{11}$

(c) $\frac{1}{2} - \frac{2}{7}$

(d) $\frac{3}{4} - \frac{1}{9}$

(e) $\frac{1}{3} - \frac{2}{9}$

(f) $\frac{5}{6} - \frac{3}{4}$

(a) $\frac{1}{3} - \frac{1}{5}$

(b) $\frac{1}{3} - \frac{2}{11}$

(c) $\frac{1}{2} - \frac{2}{7}$

(d) $\frac{3}{4} - \frac{1}{9}$

(e) $\frac{1}{3} - \frac{2}{9}$

(f) $\frac{5}{6} - \frac{3}{4}$

(a) $\frac{7}{12} + \frac{2}{5}$

(b) $\frac{8}{9} - \frac{1}{4}$

(c) $\frac{5}{6} - \frac{3}{7}$

(d) $\frac{9}{20} + \frac{2}{5}$

(e) $\frac{11}{15} - \frac{1}{6}$

(f) $\frac{2}{7} + \frac{3}{8}$

(a) $\frac{7}{12} + \frac{2}{5}$

(b) $\frac{8}{9} - \frac{1}{4}$

(c) $\frac{5}{6} - \frac{3}{7}$

(d) $\frac{9}{20} + \frac{2}{5}$

(e) $\frac{11}{15} - \frac{1}{6}$

(f) $\frac{2}{7} + \frac{3}{8}$

(a) $\frac{7}{20} + \frac{2}{5} + \frac{1}{10}$

(b) $\frac{3}{8} + \frac{1}{4} + \frac{1}{6}$

(c) $\frac{7}{9} + \frac{1}{18} - \frac{2}{3}$

(d) $\frac{43}{50} - \frac{1}{5} - \frac{7}{20}$

(a) $\frac{7}{20} + \frac{2}{5} + \frac{1}{10}$

(b) $\frac{3}{8} + \frac{1}{4} + \frac{1}{6}$

(c) $\frac{7}{9} + \frac{1}{18} - \frac{2}{3}$

(d) $\frac{43}{50} - \frac{1}{5} - \frac{7}{20}$

$$5\frac{1}{3} - 3\frac{2}{5}$$

$$4\frac{3}{4} - 2\frac{5}{7}$$

$$2\frac{8}{9} + 3\frac{3}{5}$$

$$2\frac{13}{20} - \frac{7}{8}$$



Something wrong with the bill. Can you try to figure it out....

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Bill
Coffee Shop
© math-only-math.com

© math-only-math.com

Bill No. 4588
Date:

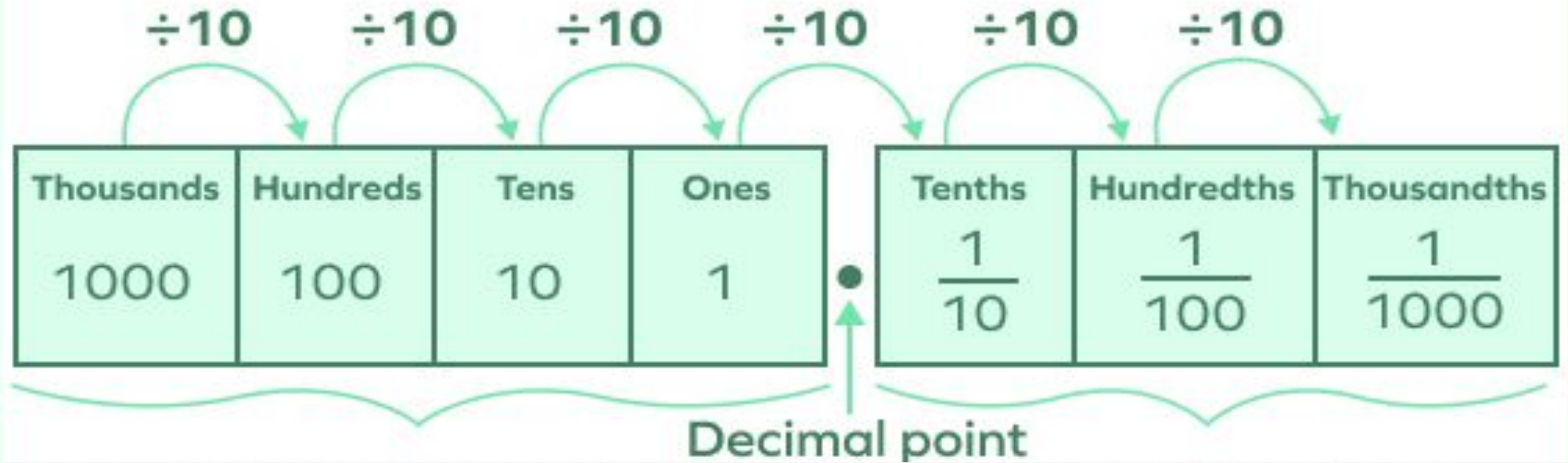
S. No.	Items	Quantity	Price (in ₹)	Amount ₹ p
1.	Hot coffee	1	12.50	12.50
2.	Club sandwich	2	30.25	60.50
3.	French fries	1	25.00	25.00
4.	Cold coffee	1	18.50	18.50
5.	Coleslaw sandwich	3	32.25	96.75
			Total	213.05

Prompts to reflect

- What do you notice?
- Which numbers catch your attention?
- What mistake might have been made?
- What does each digit after the decimal represent?
- Why is place value important?

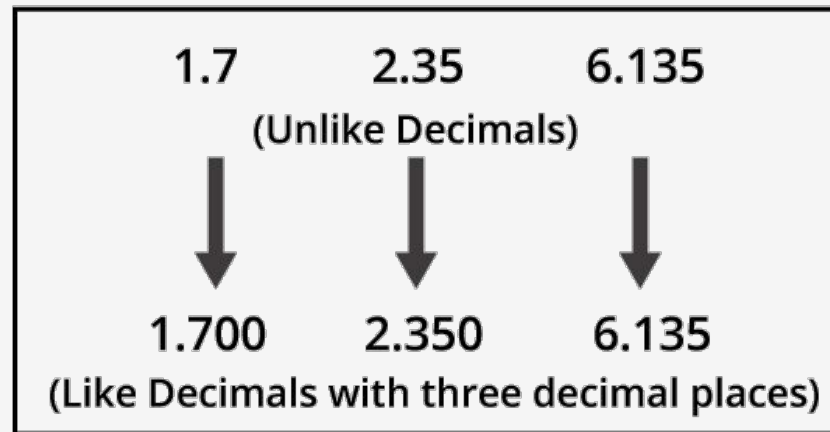
Learning Objective: Recognize and apply place value to add and subtract decimal numbers accurately.



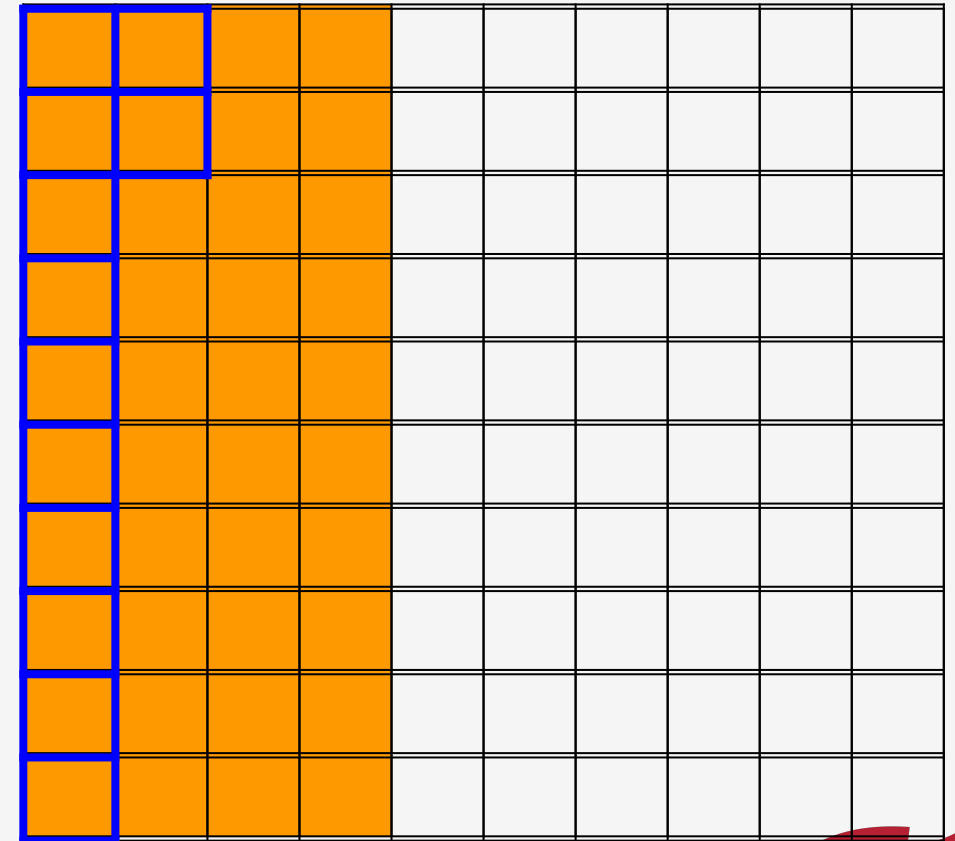
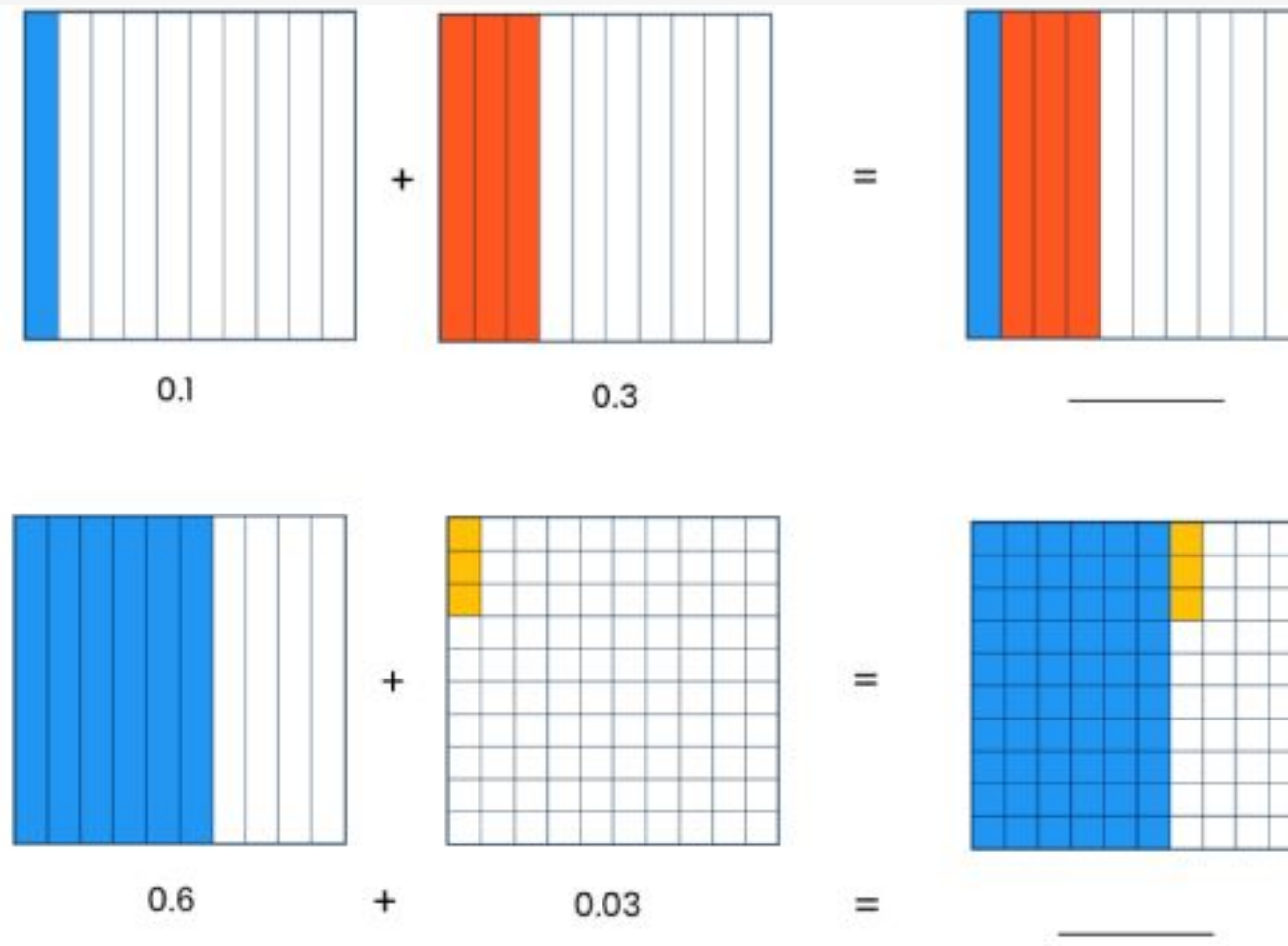


Like decimal numbers: Decimal numbers that have the same number of digits after the decimal point.

Unlike decimal numbers: Decimal numbers that have the different number of digits after the decimal point.



Addition & Subtraction of Decimals



$$0.4 - 0.12 = 0.28$$

Addition - <https://www.geogebra.org/m/umbtydf3>

Addition & Subtraction of Decimals and Whole numbers

Add 15 & 12.56

Addition

$$\begin{array}{r} 15.00 \\ + 12.56 \\ \hline 27.56 \end{array}$$

Subtract 2.25 from 6

Subtraction

$$\begin{array}{r} \overset{\textcircled{9}}{} \\ \overset{\textcircled{5}}{\cancel{6}}.\overset{\textcircled{10}}{\cancel{0}}\overset{\textcircled{10}}{\cancel{0}} \\ - 2.25 \\ \hline 3.75 \end{array}$$



Task: Find the sum & difference of the following decimals

- 1) $4.32 + 12.16 =$ _____
- 2) $12.16 - 3.14 =$ _____
- 3) $46.30 - 03.67 =$ _____
- 4) $2.1 + 21.83 =$ _____
- 5) $1.89 + 5.91 =$ _____
- 6) $23.7 + 16.1 =$ _____
- 7) $15.47 + 12.26 =$ _____
- 8) $1.7 - 18.58 =$ _____
- 9) $19.16 + 2.48 =$ _____
- 10) $8.8 - 8.8 =$ _____

① $3.897 + 1.127 =$

② $9.387 - 1.819 =$

③ $1.944 - 1.174 =$

④ $5.234 + 4.199 =$

⑤ $3.092 - 1.308 =$

⑥ $4.448 - 3.432 =$

⑦ $9.423 + 3.231 =$

⑧ $4.982 - 2.999 =$

⑨ $7.224 + 6.331 =$

⑩ $6.104 - 5.503 =$



Solve

1. $0.002 + 0.0583 =$ _____

2. $0.00016 + 0.00018 =$ _____

3. $3.940 + 0.00390 =$ _____

4. $0.00001 + 0.230 =$ _____

1. $0.600 - 0.067 =$ _____

2. $0.00900 - 0.00021 =$ _____

3. $0.0900 - 0.0012 =$ _____

4. $0.500 - 0.0098 =$ _____

Find the missing value

1) $6.4 + \underline{\hspace{1cm}} = 8.4$

2) $2.5 + \underline{\hspace{1cm}} = 8.1$

3) $7.0 + \underline{\hspace{1cm}} = 14.7$

4) $9.7 + \underline{\hspace{1cm}} = 12.4$

5) $4.9 + \underline{\hspace{1cm}} = 9.8$

6) $9.5 + \underline{\hspace{1cm}} = 18.8$



Know & Understand



A recipe requires

$\frac{8}{12}$ cup of milk.

Your measuring cup only has

$\frac{1}{3}$, $\frac{1}{2}$, $\frac{2}{3}$

Which measuring cup would you use?

Explain your reasoning.

Learning Objective: Apply knowledge of factors and multiples to simplify fractions efficiently.

Reflect:

- Can two different fractions represent the same quantity?
- Why do mathematicians simplify fractions?
- How do factors help us simplify fractions?
- How do you know when a fraction is in its simplest form?

Simplifying a Fraction: Reducing a fraction to its lowest form.

$$\frac{3}{6} \div 3 = \frac{1}{2}$$

3 and 6 share a greatest common factor of 3, so the fraction can be simplified

1 and 2 do not share a greatest common factor, so the fraction can not be simplified further

The simplest form of a fraction is equivalent to the original fraction



Example #1

Factors of 8 : 1, 2, 4, 8 →
Factors of 32 : 1, 2, 4, 8, 16, 32

$$\frac{8}{32} \div 8 = \frac{1}{4}$$

Example #2

Factors of 18 : 1, 2, 3, 6, 9, 18 →
Factors of 27 : 1, 3, 9, 27

$$\frac{18}{27} \div 9 = \frac{2}{3}$$



Using common factors also we can simplify through multiple steps.

Diagram illustrating the simplification of the fraction $\frac{18}{24}$ to $\frac{3}{4}$ through two steps:

- Step 1: $\frac{18}{24} \xrightarrow{\div 2} \frac{9}{12}$ (Dividing both numerator and denominator by 2)
- Step 2: $\frac{9}{12} \xrightarrow{\div 3} \frac{3}{4}$ (Dividing both numerator and denominator by 3)

The final simplified fraction is $\frac{3}{4}$.

Diagram illustrating the simplification of the fraction $\frac{24}{108}$ to $\frac{2}{9}$ through four steps:

- Step 1: $\frac{24}{108} \xrightarrow{\div 2} \frac{12}{54}$ (Dividing both numerator and denominator by 2)
- Step 2: $\frac{12}{54} \xrightarrow{\div 2} \frac{6}{27}$ (Dividing both numerator and denominator by 2)
- Step 3: $\frac{6}{27} \xrightarrow{\div 3} \frac{2}{9}$ (Dividing both numerator and denominator by 3)

The final simplified fraction is $\frac{2}{9}$.

Simplify the following fractions.

$\frac{4}{6} = \frac{2}{\square}$	$\frac{8}{16} = \frac{\square}{\square}$	$\frac{7}{21} = \frac{\square}{3}$	$\frac{9}{15} = \frac{\square}{\square}$
$\frac{2}{10} = \frac{\square}{\square}$	$\frac{3}{12} = \frac{\square}{\square}$	$\frac{5}{20} = \frac{\square}{\square}$	$\frac{8}{20} = \frac{\square}{\square}$
$\frac{4}{6} = \frac{2}{3}$	$\frac{12}{24} = \frac{1}{2}$	$\frac{10}{15} = \frac{2}{3}$	$\frac{4}{16} = \frac{1}{4}$
$\frac{9}{12} = \frac{\square}{\square}$	$\frac{10}{30} = \frac{\square}{\square}$	$\frac{10}{25} = \frac{\square}{\square}$	$\frac{10}{16} = \frac{\square}{\square}$

$\frac{4}{16} = \frac{\square}{\square}$	$\frac{14}{21} = \frac{\square}{\square}$	$\frac{6}{15} = \frac{\square}{\square}$	$\frac{18}{34} = \frac{\square}{\square}$
$\frac{9}{12} = \frac{\square}{\square}$	$\frac{12}{20} = \frac{\square}{\square}$	$\frac{3}{33} = \frac{\square}{\square}$	$\frac{9}{18} = \frac{\square}{\square}$
$\frac{15}{25} = \frac{\square}{\square}$	$\frac{6}{8} = \frac{\square}{\square}$	$\frac{15}{20} = \frac{\square}{\square}$	$\frac{4}{6} = \frac{\square}{\square}$
$\frac{18}{45} = \frac{\square}{\square}$	$\frac{18}{24} = \frac{\square}{\square}$	$\frac{15}{21} = \frac{\square}{\square}$	$\frac{16}{24} = \frac{\square}{\square}$



Applying Knowledge - Scenario based sums

At a birthday party, a large cake is cut into 24 equal slices. The guests eat 18 slices of the cake.

- **Task A:** Write the amount of cake eaten as a fraction of the whole cake.
- **Task B:** Simplify this fraction to find out what simplified portion of the cake was consumed.

In an MYP 1 class of 30 students, a survey shows that 12 students choose football as their favorite physical education sport.

- **Task:** Express the number of football-loving students as a simplified fraction of the whole class.

Liam has a smartphone with 64 gigabytes (GB) of total storage space. His downloadable apps and games take up 40GB of space.

- **Task:** What simplified fraction of Liam's total phone storage is used by apps and games?

A community garden covers an area of 100 square meters. The volunteers dedicate 35 square meters of the garden exclusively to planting wildflowers to save local bees.

- **Task:** What simplified fraction of the community garden is used for wildflowers?



Sports Team Selection

Students registered for

Basketball: 8 boys and 4 girls

Volleyball: 12 boys and 6 girls

Reflect

- Do the teams have the same composition?
- How can we compare fairly?
- Does having more students change the ratio?
- Where do we use ratios in everyday life?
- Why are proportions useful in recipes, maps, and construction?

Learning Objective: Compare two quantities using ratios & Proportions.



A ratio is a simple comparison between two quantities (both of them are same kind or in the same unit)

Ratios can be expressed in 3 different ways:

Ratio	Fraction	Words
2 : 3	$\frac{2}{3}$	2 to 3



RATIO AND PROPORTION

RATIO

A ratio is a way to compare two quantities.



4 : 2

2 : 1
(2 to 1)

2 : 1
(2 to 1)

PROPORTION

A proportion is an equation that shows two ratios are equal

$$\frac{4}{2} = \frac{8}{4}$$

$$2 : 1 = 2 : 1$$



Simplify the Ratios

Write the given ratios in the simplest form

3 : 15	15 : 30	6 : 36	15 : 20
= ____	= ____	= ____	= ____
12 : 36	10 : 110	9 : 27	5 : 125
= ____	= ____	= ____	= ____
25 : 50	18 : 72	8 : 30	10 : 48
= ____	= ____	= ____	= ____



Share £50 in the ratio **2:3**

- 1) Find the **total number of parts**

$$2 + 3 = \boxed{5}$$

- 2) Divide the **amount** by the **total number of parts**

$$£50 \div 5 = \boxed{£10} = 1 \text{ part}$$

- 3) Multiply each number in the **ratio** by the value of **1 part**

$$\begin{array}{c} 2:3 \\ \swarrow \quad \searrow \\ \text{x } £10 \quad \text{x } £10 \\ \hline £20:£30 \end{array}$$

Share \$48 in the ratio **3:1:2**

- 1) Find the **total number of parts**

$$3 + 1 + 2 = \boxed{6}$$

- 2) Divide the **amount** by the **total number of parts**

$$\$48 \div 6 = \boxed{\$8} = 1 \text{ part}$$

- 3) Multiply each number in the **ratio** by the value of **1 part**

$$\begin{array}{c} 3 : 1 : 2 \\ \swarrow \quad \downarrow \quad \searrow \\ \text{x } \$8 \quad \text{x } \$8 \quad \text{x } \$8 \\ \hline \$24 : \$8 : \$16 \end{array}$$

Problem Solving

2. Divide £700 in the ratio 5 : 3 : 2

3. Alex and Thomas share 30 sweets.
They divide them in the ratio 3:2.

How many sweets does Thomas have?

4. Sophie has 60 pencils.
The ratio of sharpened pencils to blunt pencils is 4:1

How many sharpened pencils does Sophie have?



Group Task: Fold the given paper and make a note.

No. of FOLDS	No. of LAYERS
0	1
1	2

Reflect using VTR -

I Notice_____

I Wonder_____

- What pattern do you notice?
- How is each number related to the previous one?
- Can you predict the number of layers after 7 folds?
- How can we represent it?

Exponents & Powers



THE GAME OF RICE

The ruler of India many years ago was delighted with the newly created game of chess. So pleased in fact that he wanted to reward its creator, who as it just so happens was somewhat of a wise man and full on mathematician.

The wise man appeared humble when he asked for simply 1 grain of rice to be placed on the first square of the chessboard on the first day, then double that number of grains of rice on the second square on the next day, and so on until all 64 squares had been filled.

It all sounded very simple and cheap and the ruler readily agreed. And sure enough it set off simple enough.

Square 1, 1 grain

Square 2, 2 grains

Square 3, 4 grains

This wise man's a fool thought the ruler.

Square 20, 524,288 grains

Then along comes **day 32** and it's an amazing **2,147,483,648**

And then something quite dreadful happens, it's at this halfway point of doubling that this

That wise man was no fool.

In fact the phrase **the second half of the chessboard** is often used to explain how quickly growth comes to have a significant impact



Find prime factorisation of 128.

We see that

$$128 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$$

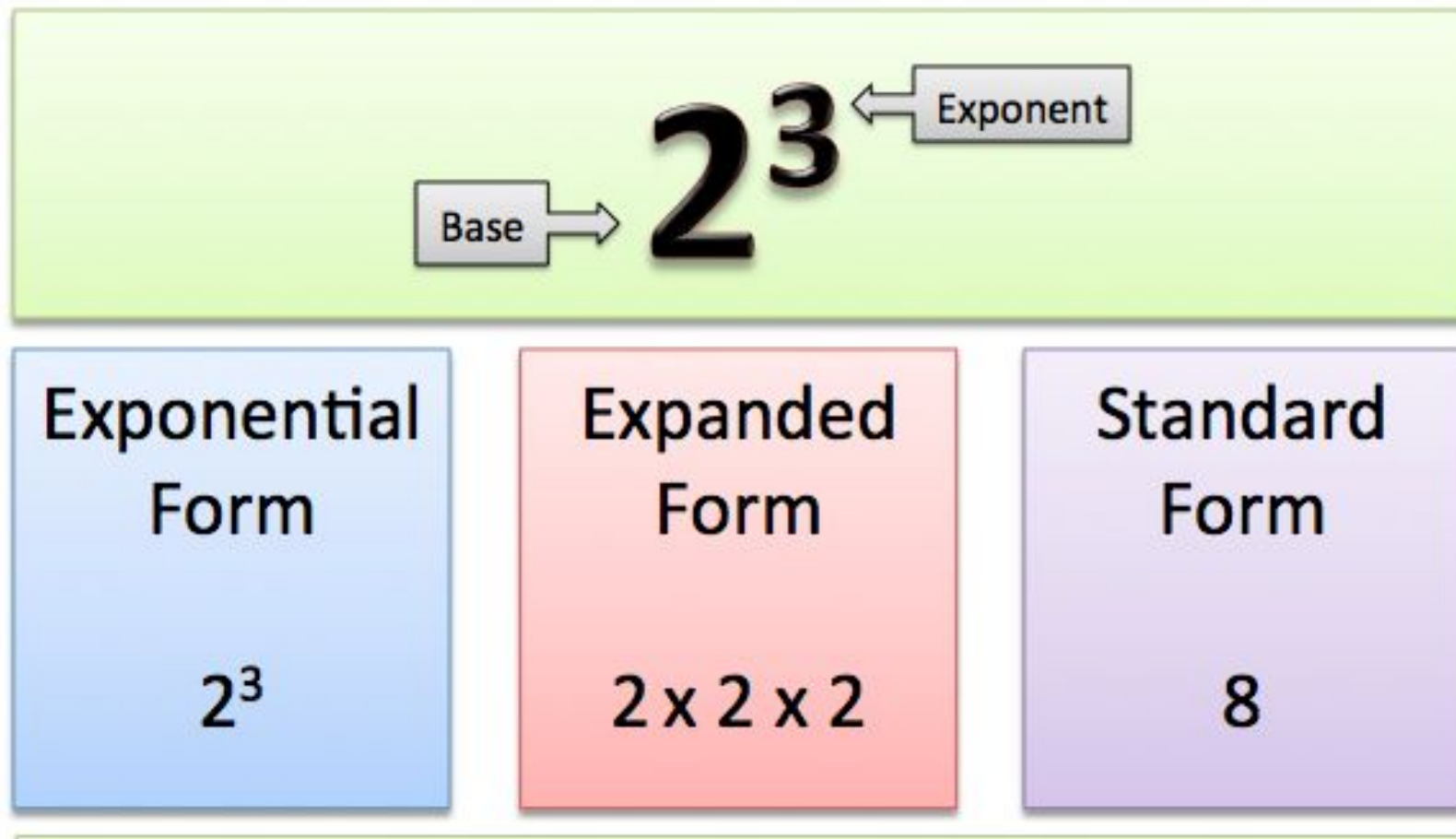
2	128
2	64
2	32
2	16
2	8
2	4
2	2
	1

This can be written as..... in Exponential form.

$$128 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2$$

$$128 = 2^7$$





An exponent tells you how many times a base number is multiplied by itself.

An exponential expression consists of two main parts:

the **base** (the number being multiplied)

the **exponent** (the small raised number telling you how many times).

Rewrite the following exponents using repeated multiplication and solve.

For example, $3^2 = 3 \times 3 = 9$

1) $11^2 =$ _____

2) $7^2 =$ _____

3) $3^4 =$ _____

4) $9^2 =$ _____

5) $2^3 =$ _____

6) $10^4 =$ _____

7) $4^6 =$ _____

8) $6^3 =$ _____

9) $8^4 =$ _____

10) $5^5 =$ _____

Rewrite the following expressions using exponents.

For example, $5 \times 5 = 5^2$

11) $5 \times 5 \times 5 \times 5 \times 5 =$ _____

12) $6 \times 6 \times 6 =$ _____

13) $18 \times 18 =$ _____

14) $4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 \times 4 =$ _____

15) $3 \times 3 \times 3 \times 3 \times 3 =$ _____

16) $9 \times 9 \times 9 =$ _____

17) $7 \times 7 \times 7 \times 7 =$ _____

18) $10 \times 10 \times 10 \times 10 \times 10 =$ _____

19) $11 \times 11 \times 11 =$ _____

20) $20 \times 20 =$ _____

Write the given numbers as exponents.

- a) 18
- b) 72
- c) 100
- d) 243
- e) 625

Square of a Number

When an integer is multiplied by itself, it is called the square of that number.

- $5^2 = 5 \times 5 = 25$
- $12^2 = 12 \times 12 = 144$

Cube of a Number

When we multiply an integer by itself three times, it is called a cube of that number.

- $2^3 = 8$
- $11^3 = 1331$



Write the Squares of first 20 natural numbers & Cubes of first 10 natural numbers.

$$1^2 = 1$$

$$2^2 = 4$$

$$3^2 = 9$$

$$4^2 = 16$$

$$5^2 = 25$$

$$6^2 = 36$$

$$7^2 = 49$$

$$8^2 = 64$$

$$9^2 = 81$$

$$10^2 = 100$$

$$11^2 = 121$$

$$12^2 = 144$$

$$13^2 = 169$$

$$14^2 = 196$$

$$15^2 = 225$$

$$16^2 = 256$$

$$17^2 = 289$$

$$18^2 = 324$$

$$19^2 = 361$$

$$20^2 = 400$$

$$1^3 = 1$$

$$2^3 = 8$$

$$3^3 = 27$$

$$4^3 = 64$$

$$5^3 = 125$$

$$6^3 = 216$$

$$7^3 = 343$$

$$8^3 = 512$$

$$9^3 = 729$$

$$10^3 = 1000$$

$$11^3 = 1331$$

$$12^3 = 1728$$

$$13^3 = 2197$$

$$14^3 = 2744$$

$$15^3 = 3375$$

$$16^3 = 4096$$

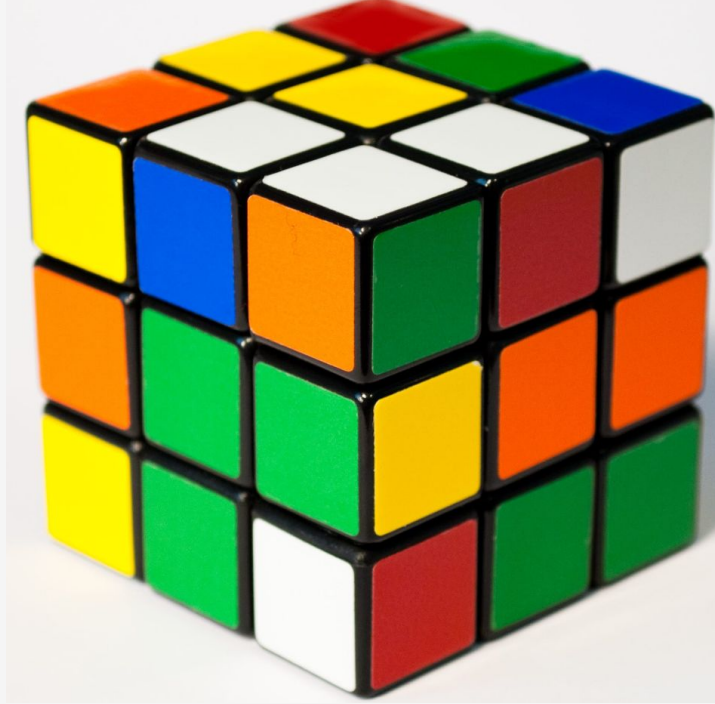
$$17^3 = 4913$$

$$18^3 = 5832$$

$$19^3 = 6859$$

$$20^3 = 8000$$





- Which objects represent squares?
- Which represent cubes?
- If you know the total number of small pieces, how could you find the side or edge length?
- Where do you see square numbers or cube numbers in everyday life?



Task: Spot the Odd One Out

9, 16, 20, 64

8, 27, 64, 45

Which number is different?
Explain your thinking.

$$\begin{aligned}\square \times \square &= 9 \\ \square \times \square &= 16 \\ \square \times \square &= 64\end{aligned}$$

$$\begin{aligned}\square \times \square \times \square &= 8 \\ \square \times \square \times \square &= 27 \\ \square \times \square \times \square &= 64\end{aligned}$$

Inverse Operations:

$+$ and $-$

$\times$ and $\div$

$\sqrt{\quad}$ and $\square^2$



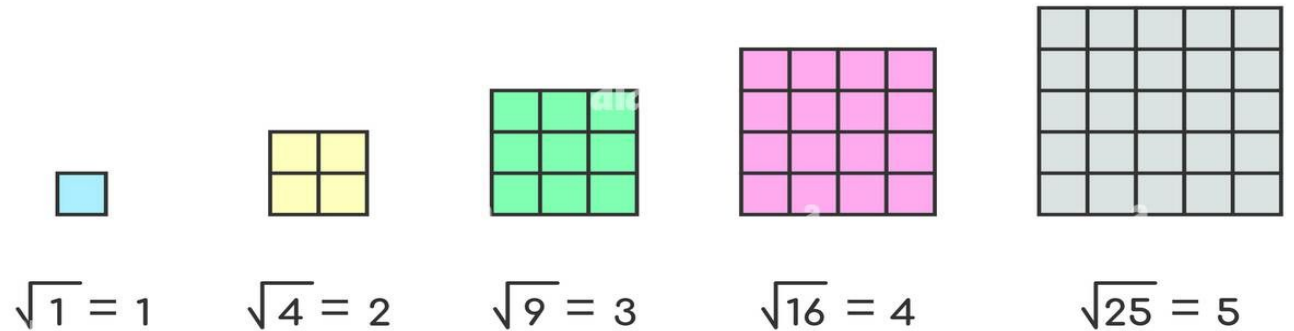
Square Root

Square Root: A value that, when multiplied by itself, gives the original number.

- Symbol: $\sqrt{\quad}$

index 2 *radical symbol* $\sqrt{\quad}$ x $= z$ $\leftarrow$ *root* *radicand*

SQUARE ROOT



Cube root

A value that, when multiplied by itself three times, gives the original number.

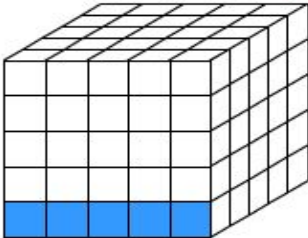
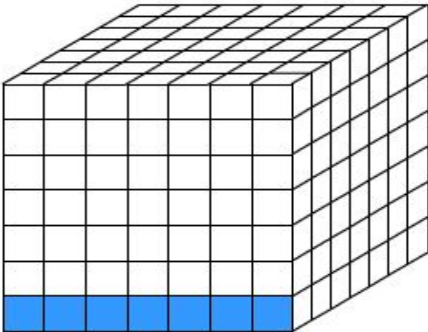
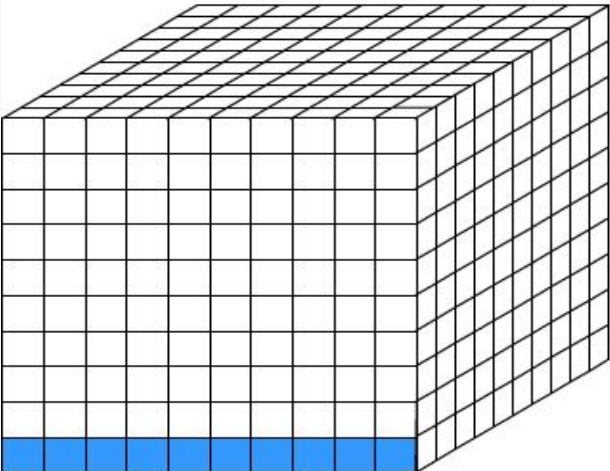
Symbol: $\sqrt[3]{}$

index → $\sqrt[3]{x} = z \leftarrow \text{root}$

radical symbol → $\sqrt{}$

radicand → x

Symbol $\sqrt[3]{}$

	$5 \times 5 \times 5 = 125$ so $\sqrt[3]{125} = 5$
	$7 \times 7 \times 7 = 343$ so $\sqrt[3]{343} = 7$
	$10 \times 10 \times 10 = 1000$ so $\sqrt[3]{1000} = 10$

Find the answer.

11) $\sqrt[2]{16} = 4$

12) $\sqrt[2]{25} =$

13) $\sqrt[2]{100} =$

14) $\sqrt[2]{81} =$

15) $\sqrt[2]{144} =$

16) $\sqrt[2]{1600} =$

17) $\sqrt[2]{8100} =$

18) $\sqrt[2]{49} =$

19) $\sqrt[2]{121} =$

20) $\sqrt[2]{64} =$



Find the answer.

21) $\sqrt[3]{8} =$

22) $\sqrt[3]{27} =$

23) $\sqrt[3]{125} =$

24) $\sqrt[3]{1000} =$

25) $\sqrt[3]{512} =$

26) $\sqrt[3]{125} =$

27) $\sqrt[3]{64} =$

28) $\sqrt[3]{343} =$

29) $\sqrt[3]{729} =$

30) $\sqrt[3]{216} =$



The Digit Investigation

- Look at the final digit of the first ten perfect squares: 1, 4, 9, 16, 25, 36, 49, 64, 81, 100.
- List all the digits that a perfect square **can** end with.
- List the digits that a perfect square **can never** end with.
- **Predict:** Based on your rule, could the number 2,348 be a perfect square? Explain.



Task: Applied Questions

- A square garden has an area of 81 Sqm What is the length of one side?
- A metal cube has a volume of 64 Cubic cm. What is the length of one edge?
- A storage box shaped like a cube has a volume of 216 Cubic cm. What is the height of the box?
- **Reasoning:** Maya says that the square root of a number is always smaller than the number itself. Is she correct? Give an example to prove your point.

